

Sangram Kumar Sahoo

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EDUCATION

MS in Mechanical Engineering - Carnegie Mellon University (CMU)

Expected Graduation: May 2026 | GPA 4.0/4.0

Subjects: MLAI for Engineers, Modern Control Theory, Soft Robots, Intro to Deep Learning, Engineering Startup, Computer Vision, Deep Reinforcement Learning

B.Tech in Mechanical Engineering – SRM Institute of Science and Technology, Chennai

Graduated: May 2023 | GPA: 8.7/10

PROJECTS

Detection of water purity level using an underwater drone system

(Aug - Dec 2024)

- The technical problem this project aims to address is the development of a bio-inspired soft robotic device that monitors water quality and detects hydrogen gas in water bodies, overcoming the limitations of stationary sensors and manual sampling methods. Soft robotic structures affect maneuverability and the impact of sensor placement on data accuracy in dynamic water conditions.

Reinforcement Learning-based neural architecture search for deepfake voice detection

(Jan - May 2025)

- Established a comprehensive data processing pipeline for audio preprocessing and feature extraction
- Implemented and evaluated baseline models, achieving an Equal Error Rate (EER) of 2.8% with 50 epochs
- Developed an initial proof-of-concept for the Neural Architecture Search framework (currently under refinement)

PROFESSIONAL EXPERIENCE

Computational Engineering and Robotics Lab (CERLAB)

CMU, Pittsburgh

Graduate Researcher - RoboConstruction Team.

(Aug 2024 -May 2025)

- Developed an Isaac Sim simulation environment for an SLAM test bed with the ultimate goal of creating a SLAM algorithm for a mining environment.
- Simulated different LiDARs and camera integration for straightforward hauling and loading operations for the wheel loader in a simulated environment.

Graduate Researcher - Shapeanalysis Team.

(May 2025 -Present)

- My work focuses on enhancing the representation and analysis of complex shapes by leveraging hierarchical structures for clarity and organization. I utilize dynamic graph models for efficient feature extraction and actively master key algorithms to improve my analytical capabilities.

Kansai Nerolac Paints

Hosur, India

Mechanical Engineering Intern

(Jun - Jun 2022)

- Completed the daily production checklist and generated comprehensive reports on daily activities. Operated industrial machinery and acquired a thorough understanding of various manufacturing processes. Assisted in maintaining the ball mill and the daily routine of the RO water system within the manufacturing plant.

Youth India E-school

Bhubaneswar, India

Deputy General Manager

(Dec - July 2024)

- Oversee all activities related to operational metrics, quality standards, and human resources management. Oversee and train four sales associates, two team leaders, and two support managers. Provide employees with training in customer service and conflict-resolution techniques.

RESEARCH EXPERIENCE

Design of pedicle screw fixation for lumbar spine

(Jan – May 2023)

SRM Institute of science and technology

- Developed an FE model to study pedicle screw fixation and the effect of screws by varying the size and material. The distribution of stress across different screws with varying forces applied to the vertebrae was observed.

SKILLS

Frameworks: Python (PyTorch, TensorFlow, CUDA, PySpark, OpenCV), C++, Docker, ROS, Kotlin.

Database & Cloud: Google Cloud Platform (GCP), AWS, Power BI.

Software: Solidworks, Autocad, Matlab

LEADERSHIP

- **Youth India Foundation**, Co-Founder

Feb 2020 – Present